

Fixed-Parameter Additive Gaps for Deterministic Communication Complexity

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Abstract

For every fixed positive integer k , we construct, from an ordered 3-CNF formula Φ , a nonempty family of Boolean matrices and integer budgets. The family is chosen without knowing whether Φ is satisfiable. If Φ is satisfiable, at least one matrix has deterministic communication complexity at most its budget. If Φ is unsatisfiable, every matrix has complexity exactly its budget plus k . The same matrices have at most $2^{\lceil n/(2k) \rceil}$ rows and columns, and hence at most $2^{\lceil n/k \rceil}$ entries, once the number n of variables exceeds a cutoff depending only on k .

The proof is relative to two explicit finite inputs: a composed source theorem and a balanced-family theorem. Starting from a robust full-cube matrix, one level of the construction runs the first three stages of the parameterised hardness gadget, forms a raw fourth stage, protects it by labelled duplication, and then aligns two different lower-bound scales by an interlace and transpose. A construction-independent bridge adds exactly one bit while restoring the robust property needed at the next level. Iterating the same promise-independent transition k times gives the gap. Exact carrier identities for that very matrix recursion give the size bound.

1 The finite gap theorem

Alice receives a row of a finite Boolean matrix M , Bob receives a column, and they must output the corresponding entry by a deterministic binary protocol. We write $D(M)$ for the minimum worst-case number of communicated bits [3, 4]. All matrices whose communication complexity is discussed have nonempty finite row and column sets.

The input to our construction is an *ordered* 3-CNF formula Φ : the clause order, the three positions within each clause, and repeated literals are part of the input. Let $n(\Phi)$ be its declared number of variables. For $k \geq 1$, put

$$e(n, k) = \left\lfloor \frac{n}{2k} \right\rfloor, \quad S(n, k) = 2^{e(n, k)}, \quad E(n, k) = S(n, k)^2. \quad (1)$$

For a nonempty finite matrix define its *naming width* by

$$w(M) = \max \{1, \lceil \log_2 |\text{Rows}(M)| \rceil, \lceil \log_2 |\text{Cols}(M)| \rceil\}. \quad (2)$$

The ceiling logarithms are natural-number ceiling logarithms, so the definition also covers a one-element carrier.

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Theorem 1.1 (Finite fixed-parameter gap and same-matrix size). *Fix a composed finite source theorem and a balanced-family theorem satisfying the two contracts in [Section 2.2](#). There is one promise-independent construction with the following property. For every $k \geq 1$ there is an integer $U_k \geq k$, chosen before the formula, such that every valid ordered 3-CNF formula Φ with $n(\Phi) \geq U_k$ produces a positive integer $t(\Phi)$ and a family*

$$(M_{\Phi,k,j}, Y_{\Phi,k,j}), \quad j \in [t(\Phi)].$$

Every matrix in the family has nonempty row and column sets, and the entire family is fixed before satisfiability is used. Moreover,

$$\Phi \text{ satisfiable} \implies \exists j \in [t(\Phi)] \quad D(M_{\Phi,k,j}) \leq Y_{\Phi,k,j}, \quad (3)$$

$$\Phi \text{ unsatisfiable} \implies \forall j \in [t(\Phi)] \quad D(M_{\Phi,k,j}) = Y_{\Phi,k,j} + k, \quad (4)$$

$$\forall j \in [t(\Phi)] \quad w(M_{\Phi,k,j}) \leq e(n(\Phi), k). \quad (5)$$

For those same matrices,

$$|\text{Rows}(M_{\Phi,k,j})| \leq S(n(\Phi), k), \quad |\text{Cols}(M_{\Phi,k,j})| \leq S(n(\Phi), k), \quad (6)$$

$$|\text{Rows}(M_{\Phi,k,j})| |\text{Cols}(M_{\Phi,k,j})| \leq E(n(\Phi), k) \leq 2^{\lfloor n(\Phi)/k \rfloor}. \quad (7)$$

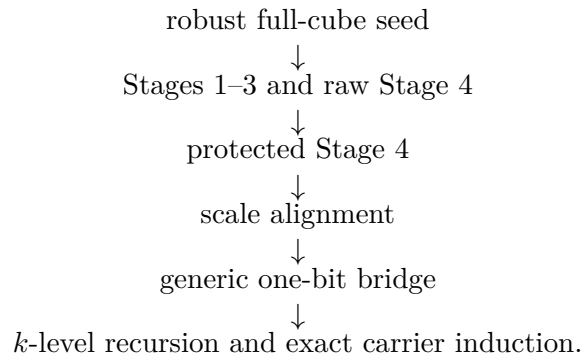
Three parts of the statement are worth making explicit. The cutoff U_k does not depend on the formula or on a source branch. The matrices are not chosen differently in the two formula cases. Finally, the size bounds apply to the matrices in the communication gap, not to padded or replacement objects.

1.1 Why scale alignment is necessary

The one-bit fourth-stage gadget inherited from the exact hardness proof is not directly iterable. Iteration needs two lower bounds at a common small row density, while the raw matrix has no full-cost bound of that kind. There are two simple obstructions. An off-diagonal column restriction retains at least 7/8 of the columns while making all active source entries neutral. Independently, if an optimal protocol begins with Alice, one child keeps at least half the rows and has residual cost at most one bit below the original cost. Thus the tempting direct low-density full-cost anchor is false.

The proof makes two repairs. Labelled duplication first creates a protected matrix N with a near-full full-cost estimate and with three one-bit-down estimates on an actual reference restriction. An interlace followed by transposition then creates W , where these estimates share one threshold. A generic bridge turns W into the next robust seed and adds exactly one bit.

The complete proof has the following form:



The presentation below compresses the formalisation-facing manuscript but does not change this route. In particular, it uses the same external inputs, the same matrices, the same positive and negative budgets, and the same carrier recurrence.

2 Finite communication and the two external inputs

For $m \in \mathbb{N}$, write $[m] = \{0, \dots, m-1\}$. A restriction $M|_{A \times B}$ uses the literal subtype row and column sets. Whenever we take its communication complexity we first prove A and B nonempty. Transposition and bijective relabelling preserve communication complexity; restriction cannot increase it; and duplicating equal rows or columns preserves it if one representative of every original input remains.

We use one elementary lower bound repeatedly. In a reduced depth- d protocol for a nonconstant Boolean matrix, at most 2^{d-1} leaves have any fixed output bit. The entries reaching each such leaf form a rectangle, so if the real rank of the indicator matrix of one output exceeds 2^{d-1} , then the communication complexity is at least $d+1$.

2.1 Interlaces and retained profiles

For $p \geq 1$ and $M : X \times Y \rightarrow \{0, 1\}$, define

$$\text{Int}_p(M)((i, x), f) = M(x, f(i)), \quad (i, x) \in [p] \times X, \quad f : [p] \rightarrow Y. \quad (8)$$

Its two carrier sizes are

$$|\text{Rows}(\text{Int}_p(M))| = p|X|, \quad |\text{Cols}(\text{Int}_p(M))| = |Y|^p. \quad (9)$$

If p is a positive power of two, the parties can first communicate the coordinate i and then run a protocol for M , giving

$$D(\text{Int}_p(M)) \leq D(M) + \log_2 p. \quad (10)$$

To state the lower bounds, let $\mathcal{D}_M(p; x, z)$ be the minimum communication complexity among restrictions of $\text{Int}_p(M)$ that retain at least $\lceil x|X| \rceil$ rows in *each* outer copy and at least $\lceil z|Y|^p \rceil$ columns. The row threshold is per copy, not merely a total threshold. Define

$$\Lambda_M(p; x, z) = \min \{ \mathcal{D}_M(p; x, z), 1 + \mathcal{D}_M(p; x, z/2), 2 + \mathcal{D}_M(p; x, z/4) \} \quad (11)$$

and

$$\text{Rob}(M; b, z) = \Lambda_M(1; 2^{-b}, z). \quad (12)$$

Thus $C \leq \text{Rob}(M; b, 3/5)$ simultaneously says that the three retained-column densities $3/5, 3/10, 3/20$ have lower bounds $C, C-1, C-2$. Bijections of both carriers preserve $\mathcal{D}$, Λ , and Rob .

2.2 The composed source theorem

An ordered multigraph G has ordered vertex and edge-occurrence carriers, with endpoint maps $\partial_0, \partial_1 : [e(G)] \rightarrow [v(G)]$. Parallel edge occurrences remain distinct. For a colouring $c : [v(G)] \rightarrow [3]$, let $\text{mono}(G, c)$ count monochromatic edge occurrences.

The first external input supplies fixed integers $c_{\text{src}}, \Delta, n_{\text{src}} \geq 1$, a fixed rational $0 < \gamma \leq 1$, and, for every ordered formula Φ , one finite ordered list $\mathcal{S}(\Phi)$ of graphs. For every valid Φ with $n = n(\Phi) \geq n_{\text{src}}$:

- (S1) the list has positive length $t(\Phi)$;
- (S2) every listed graph is nonempty, loop-free, and has every degree between 1 and Δ ;
- (S3) every listed graph has at least n edge occurrences and satisfies
$$v(G) + e(G) \leq n(\log_2(n+2))^{c_{\text{src}}};$$
- (S4) if Φ is satisfiable, some listed graph is properly three-colourable;
- (S5) if Φ is unsatisfiable, then for every listed graph and every three-colouring c , $\gamma e(G) \leq \text{mono}(G, c)$.

The same ordered list occurs in all five clauses. This is a composed finite source theorem. A proof of it may use an expander theorem, but after the list and these properties are supplied, no expander object or parameter is an input to the matrix construction. The intended source is the companion construction [2].

The use of an integer logarithm in Lean costs only a change of fixed constants. For $n \geq 2$, put $L = \lfloor \log_2(n+2) \rfloor \geq 2$. Then $\log_2(n+2) < L+1 \leq L^2$. Replacing c_{src} by $2c_{\text{src}}$ and n_{src} by $\max\{n_{\text{src}}, 2\}$ therefore yields the integer-logarithm bound $v(G) + e(G) \leq nL^{2c_{\text{src}}}$, with the original exponent on the right. The ordered graph lists, γ , Δ , and both promise claims are unchanged. We make this replacement before selecting parameters and reuse the names $c_{\text{src}}, n_{\text{src}}$ for the normalized constants.

2.3 The balanced-family theorem

For $m > 0$ and $1 \leq t \leq q$, an indexed family is $\mathcal{F} = (f_\ell : [q] \rightarrow [m])_{\ell \in [N]}$. Given a set of coordinates $J \subseteq [q]$ and a pattern $a : J \rightarrow [m]$, write

$$C_{\mathcal{F}}(J, a) = |\{\ell \in [N] : f_\ell|_J = a\}|.$$

The second external input supplies one such family for every legal (q, t, m) , with $N > 0$, all entries in $[m]$, and

$$N < (2 \max\{q, 1000mt^2, 2\})^t, \tag{13}$$

$$(100t-1)N \leq 100t m^{|J|} C_{\mathcal{F}}(J, a) \leq (100t+1)N \quad (|J| \leq t). \tag{14}$$

Such finite families are standard descendants of small almost-independent sample spaces [1]. The explicit construction below realizes both this contract and the product estimates needed next.

The scale-alignment proof also needs three explicitly named product instances. For a family $\mathcal{C} = (c_\ell : [Q] \rightarrow \mathcal{Y})_{\ell \in [L]}$, let $\text{Bal}_t(\mathcal{C}; \varepsilon)$ mean that $L > 0$ and, for every $J \subseteq [Q]$ with $r = |J| \leq t$ and every total pattern $a : [Q] \rightarrow \mathcal{Y}$,

$$\left| \frac{|\{\ell : c_\ell(\alpha) = a(\alpha) \text{ for all } \alpha \in J\}|}{L} - \frac{1}{|\mathcal{Y}|^r} \right| \leq \frac{\varepsilon}{|\mathcal{Y}|^r}. \tag{15}$$

For a finite donor set D , the literal product family $\mathcal{C}^{\times D}$ is indexed by maps $\lambda : D \rightarrow [L]$, has coordinates $(d, \alpha) \in D \times [Q]$, and has entry

$$c_{\lambda}^{\times D}(d, \alpha) = c_{\lambda(d)}(\alpha). \tag{16}$$

At one canonical level, let $\mathcal{C}$ be the actual Stage-2 family row, with parameters q_2, t_2, N_2 and alphabet X_1 . Put

$$\varepsilon_2 = \frac{1}{100t_2}, \quad \varepsilon_{\text{prod}} = \max \left\{ (1 + \varepsilon_2)^{t_2} - 1, 1 - (1 - \varepsilon_2)^{t_2} \right\}. \tag{17}$$

The additional product contract has exactly these three fields:

- (B1) for every $e \geq 1$ and every $J' \subseteq [2^e]$ with $|J'| = 2^{e-1}$, the product over the donor subtype $D_{e,J'} = J'$ satisfies $\text{Bal}_{t_2}(\mathcal{C}^{\times D_{e,J'}}; \varepsilon_2)$;
- (B2) the two-donor product satisfies $\text{Bal}_{t_2}(\mathcal{C}^{\times [2]}; \varepsilon_{\text{prod}})$;
- (B3) for the canonical exceptional exponent $e_{\text{exc}} \geq 2$, every $J' \subseteq [2^{e_{\text{exc}}}]$ with $|J'| = 2^{e_{\text{exc}}-1}$ again satisfies $\text{Bal}_{t_2}(\mathcal{C}^{\times D_{e_{\text{exc}},J'}}; \varepsilon_2)$.

These are finite counting statements about the actual family. They contain no protocol or communication lower bound. The source list and all balanced families are fixed before either formula promise is invoked.

Why one family satisfies all these estimates. Let $X = \max\{q, 1000mt^2, 2\}$ and choose the least prime p with $X < p < 2X$, which exists by Bertrand's postulate. Take all p^t polynomials of degree less than t over $\mathbb{F}_p$, evaluate them at q distinct residues, and map each value's representative in $[p]$ to its residue modulo m . This gives exactly the required size bound. Each alphabet bucket has size $b_a \in \{\lfloor p/m \rfloor, \lceil p/m \rceil\}$, so $|mb_a/p - 1| \leq \delta = m/p < 1/(1000t^2)$.

Polynomial interpolation makes any $s \leq t$ distinct field evaluations independent and uniform. In an arbitrary donor product, a pattern on $r \leq t$ distinct donor-coordinate pairs therefore has normalized count

$$\frac{m^r C(J, a)}{(p^t)^{|D|}} = \prod_{(d,j) \in J} \frac{mb_{a(d,j)}}{p} \in [(1 - \delta)^r, (1 + \delta)^r].$$

Only r , not the number of donors, controls the error. Indeed, $(1 - \delta)^r \geq 1 - r\delta$, while the binomial expansion gives $(1 + \delta)^r \leq \sum_{i=0}^r (r\delta)^i \leq (1 - r\delta)^{-1}$. Both errors are at most $1/(100t)$, since $r\delta \leq 1/(1000t)$ and $r\delta/(1 - r\delta) \leq 1/(1000t - 1) \leq 1/(100t)$. Empty patterns and empty donor sets have normalized count 1. This proves (B1) and (B3) for the same chosen family; (B2) follows because $\varepsilon_{\text{prod}} \geq \varepsilon_2$ by Bernoulli's inequality. Relabelling the alphabet and coordinate carriers changes none of these counts. For invalid parameter requests a singleton all-zero family completes the provider to a total function; no valid construction uses that fallback. The existence proof remains external to Lean. Importantly, it uses the specified polynomial family: the coarse base estimate alone would not justify retaining the same error under products of an arbitrary family.

3 One promise-independent level

Fix a source branch $S = (G, \gamma, \Delta)$ and a current seed matrix $R : [m_R] \times [n_R] \rightarrow \{0, 1\}$. One level reads six natural parameters

$$(a, s, q_2, t_2, q_{\text{al}}, Q_{\text{br}}). \tag{18}$$

The first four control the core gadget, q_{al} controls scale alignment, and Q_{br} controls the final bridge. These widths are not silently identified.

The structural assumptions are: $v(G), m_R, n_R > 0$; $a \geq 2$ and $s \geq 4$;

$$q_2 = 2^{\lceil \log_2(12se(G)) \rceil}, \quad 1 \leq t_2, \quad 4t_2 \leq q_2; \tag{19}$$

every active source item has rank below s in its guarded coordinate; and $q_{\text{al}}, Q_{\text{br}} > 0$. These are structural conditions only. They do not contain a protocol lower bound or a formula promise. If they fail, the constructor returns a fixed all-false matrix on its already determined carriers. The eventual cutoff proves that every level used by the theorem takes the genuine branch.

3.1 Stages 1–3

Put $q_1 = 2^a - 2$ and $L_1 = q_1 + s + 1$. The first stage is

$$\begin{aligned} X_1 &= [q_1] \times [m_R], & Y_1 &= ([L_1] \rightarrow [n_R]), \\ M_1((i, x), \Gamma) &= R(x, \Gamma(i)). \end{aligned} \tag{20}$$

Identify $[q_1 m_R]$ with X_1 by quotient and remainder. From the balanced family $\mathcal{F}(q_2, t_2, q_1 m_R)$, of size N_2 , obtain entries $c_{\ell, \alpha} \in X_1$. Then

$$X_2 = [q_2] \times Y_1, \quad Y_2 = [N_2], \quad M_2((\alpha, \Gamma), \ell) = M_1(c_{\ell, \alpha}, \Gamma), \tag{21}$$

$$X_3 = [s] \times Y_2, \quad Y_3 = ([s] \rightarrow X_2), \quad M_3((p, \ell), d) = M_2(d(p), \ell). \tag{22}$$

Thus the third stage is the s -copy interlace of M_2^T .

3.2 Guarded source rows and raw Stage 4

Let $V = [v(G)]$ and $E = [e(G)]$. A source item is either a vertex or one of $s - 3$ blockers, and a local coordinate is either an edge occurrence or one of its two incidence positions:

$$\mathcal{I} = V \sqcup [s - 3], \quad \mathcal{Z} = E \sqcup (E \times [2]).$$

There are $4s$ labelled slabs. Use the exact local ranks

$$\text{rank}_{\mathcal{Z}}(e) = e, \quad \text{rank}_{\mathcal{Z}}(e, b) = e(G) + 2e + b.$$

A vertex item accepts the edge coordinates on which it is an endpoint and the incidence coordinate that names it; a blocker accepts every local coordinate. Slab labels must agree. The local coordinate z in slab h is encoded by $h(3e(G)) + \text{rank}_{\mathcal{Z}}(z) \in [q_2]$. The resulting local relation gives a guarded source matrix

$$G_{\text{src}} : ([4s] \times \mathcal{I}) \times [q_2] \rightarrow \{0, 1\}. \tag{23}$$

Within each coordinate, active source rows are ordered by their ordinary zero-based rank. The structural assumptions place this rank in $[s]$.

The raw fourth-stage carriers are

$$\begin{aligned} X_B &= ([s] \times Y_2) \sqcup (([4s] \times \mathcal{I}) \times [m_R]), \\ Y_B &= ([s + 1] \rightarrow [n_R]) \times ([s] \rightarrow X_2). \end{aligned} \tag{24}$$

Write a raw column as (β, d) , and let α be the Stage-2 coordinate advertised at its first selector position. Template rows copy M_2 :

$$B((p, \ell), (\beta, d)) = M_2(d(p), \ell). \tag{25}$$

For a source item (h, i) and seed row x , if all $d(p)$ advertise the same α and $G_{\text{src}}((h, i), \alpha) = 1$, the source row outputs $R(x, \beta(r_\alpha(h, i)))$, where $r_\alpha(h, i) \in [s]$ is its active rank. Otherwise it outputs the neutral value $R(x, \beta(s))$. This is a total entry rule fixed before the graph is known to be colourable or far from colourable.

3.3 Labelled protection and the actual post-composition

Four row reservoirs and three column reservoirs of B are duplicated. Let

$$\begin{aligned} T &= [s] \times Y_2, & V_S &= ([4s] \times V) \times [m_R], & B_S &= ([4s] \times [s-3]) \times [m_R], \\ S_C &= ([s] \rightarrow X_2), & D_C &= [q_2] \times Y_1. \end{aligned}$$

The inclusions into the raw carriers are fixed. The three row reservoirs use their displayed summands of X_B ; a selector $d \in S_C$ is paired with the constant seed tuple at column 0; and $(\alpha, \Gamma) \in D_C$ is sent to the column whose seed tuple is the tail of Γ and whose selector is constantly (α, Γ) . Write

$$\begin{aligned} M_0 &= |Y_B|, & S_0 &= |S_C|, & D'_0 &= |D_C|, \\ X_0 &= |X_B|, & T_0 &= |T|, & V_0 &= |V_S|, & B_0 &= |B_S|. \end{aligned}$$

The column multiplicities are

$$w_{\text{base}} = 182S_0D'_0, \quad w_{\text{sel}} = M_0D'_0, \quad w_{\text{diag}} = M_0S_0, \quad (26)$$

and the row multiplicities are

$$\begin{aligned} u_{\text{base}} &= (2^{22} - 4)T_0V_0B_0, & u_{\text{temp}} &= 2X_0V_0B_0, \\ u_{\text{vert}} &= X_0T_0B_0, & u_{\text{block}} &= X_0T_0V_0. \end{aligned} \quad (27)$$

The protected matrix A has the labelled disjoint-union carriers

$$\begin{aligned} X_A &= (X_B \times [u_{\text{base}}]) \sqcup (T \times [u_{\text{temp}}]) \sqcup (V_S \times [u_{\text{vert}}]) \sqcup (B_S \times [u_{\text{block}}]), \\ Y_A &= (Y_B \times [w_{\text{base}}]) \sqcup (S_C \times [w_{\text{sel}}]) \sqcup (D_C \times [w_{\text{diag}}]). \end{aligned} \quad (28)$$

Forgetting labels and using the fixed inclusions into X_B, Y_B defines surjections ρ, κ , and

$$A(r, c) = B(\rho(r), \kappa(c)). \quad (29)$$

Equal labels therefore give equal rows or columns; protection changes carrier multiplicities but not communication complexity.

Now set

$$N = A^\top, \quad U = \text{Int}_{q_{\text{al}}}(N), \quad W = U^\top, \quad R^+ = \text{Int}_{Q_{\text{br}}}(W)^\top. \quad (30)$$

The reference matrix N_{ref} is the restriction of N whose rows are *all* base-labelled copies $Y_B \times [w_{\text{base}}]$ and whose columns are all of X_A . Hence

$$|\text{Cols}(N_{\text{ref}})| = |\text{Cols}(N)|. \quad (31)$$

Finally, R^+ is relabelled by the canonical bijections to zero-based row and column carriers. This canonical relabelling is the unique public successor.

3.4 The protected certificate

Let $D_0 = D(R)$ on the negative source branch, and put

$$\begin{aligned} C &= D_0 + a + \log_2 q_2 + \log_2 s + 1, \\ C_W &= C + \log_2 q_{\text{al}}. \end{aligned} \quad (32)$$

The schedule fixes positive deletion parameters η_R, η_C , a reference depth b_T , and the scalar inequalities listed in [Section C](#). Its negative seed assumptions are

$$D_0 \geq 3, \quad D(R) = D_0, \quad D_0 \leq \text{Rob}(R; b_0, 3/5), \quad (33)$$

together with the common graph facts and the far-from-three-colourable source condition.

Theorem 3.1 (Protected fourth-stage certificate). *Under the structural and negative scalar conditions, the actual N and N_{ref} above satisfy*

$$C \leq \mathcal{D}_N(1; 1 - \eta_R, 1 - \eta_C), \quad (34)$$

$$D(N) = C, \quad (35)$$

$$C - 1 - r \leq \mathcal{D}_{N_{\text{ref}}}(1; 2^{-b_T}, (3/5)/2^r) \quad (r = 0, 1, 2). \quad (36)$$

The proof is summarized in [Section A](#). Its key point is literal identity: every routed row, reference row, and localized column is a member of the retained restriction of the N defined above. The dense lower bound does not assert a low-density full-cost estimate on all of N . The three low-density rungs belong to the reference restriction.

For exactness, the dense statement applied to the full matrix gives $D(N) \geq C$. Conversely, the explicit raw protocol sends the a first-stage selector bits, the $\log_2 q_2$ second-stage coordinate bits, the $\log_2 s$ third-stage selector bits, and one bit distinguishing template and source rows before running an optimal protocol for R . Duplication and transpose preserve cost, so $D(N) \leq C$.

4 Scale alignment and the one-bit bridge

The protected theorem gives two kinds of information at different row scales. Scale alignment converts them to one common threshold on the actual matrix W from [\(30\)](#).

Choose

$$\delta_U = \frac{\eta_R}{16q_{\text{al}}}, \quad Q_{\text{br}} = \text{Qwidth}(b_+, \delta_U), \quad \theta = \frac{1}{160Q_{\text{br}}}, \quad (37)$$

where $\text{Qwidth}(b, \delta)$ is the least positive power of two Q such that $80b \leq Q\delta$. The schedule ensures $\delta_U \leq 1/64$ and supplies the remaining projection and terminal gates.

Theorem 4.1 (Scale-aligned certificate). *Under the hypotheses of [Theorem 3.1](#), the actual matrix $W = \text{Int}_{q_{\text{al}}}(N)^\top$ satisfies*

$$D(W) = C_W, \quad (38)$$

$$C_W \leq \mathcal{D}_W(1; \theta, 1 - \delta_U), \quad (39)$$

$$C_W - 1 \leq \Lambda_W(1; \theta, 3/5). \quad (40)$$

If $D(N) \leq Y_N$, then

$$D(W) \leq Y_N + \log_2 q_{\text{al}}. \quad (41)$$

Equivalently before the final transpose, the following operational statement is what the proof establishes. For every $R_U \subseteq \text{Rows}(U)$ and $C_U \subseteq \text{Cols}(U)$ satisfying

$$|R_U| \geq \lceil (1 - \delta_U) |\text{Rows}(U)| \rceil, \quad |C_U| \geq \lceil \theta |\text{Cols}(U)| \rceil,$$

every protocol for $U|_{R_U \times C_U}$ has cost at least C_W . This is a statement about the literal retained game on U , not about an abstract trace.

Here is the shape of the proof. Transpose (39) back to $U = \text{Int}_{q_{\text{al}}}(N)$. A retained member of U keeps all but a δ_U fraction of its rows and a θ fraction of its columns. Summing row loss over physical coordinates gives

$$\sum_{i < q_{\text{al}}} (|\text{Rows}(N)| - |R_i|) \leq \delta_U q_{\text{al}} |\text{Rows}(N)| = \frac{\eta_R}{16} |\text{Rows}(N)|. \quad (42)$$

The same aggregate bound controls loss from the actual reference fibres. Elementary product-projection estimates remove a bounded exceptional set of coordinates.

Assume a protocol of cost at most $C_W - 1$. Follow its actual Alice questions until Bob first speaks or until $\log_2 q_{\text{al}}$ Alice answers have been seen. Early leaves receive only a virtual route label; no query is added and the leaf remains unchanged. The resulting stopping routes are prefix-free, their row sets partition the retained rows, and the remaining subprotocol at a route of length d has cost at most $C_W - 1 - d$.

Reference-row certificates are routed separately from physical protected-row mass. A small first-Bob node is contradicted by one of the three actual reference rungs. A heavy first-Bob node is contradicted by the first product balancedness clause. At the final Alice routes, each clean reference certificate has a distinct owner route. Most of its physical coordinate's protected mass must leave that owner. The two-donor product clause shows that only a small amount can reach another good owner. Hence substantial mass reaches exceptional final routes. A double pigeonhole argument and the third product clause produce a hard restriction inside one such route, again contradicting its remaining cost. Appendix A gives the full owner-donor count.

Minimizing over retained members proves (39). The full matrix gives $D(W) \geq C_W$, while the selector protocol and (35) give the reverse inequality. Applying the three reference rungs coordinatewise gives (40). The same selector protocol proves (41) from any upper bound on N .

4.1 A construction-independent bridge

Call the three statements

$$D(W) = C_W, \quad C_W \leq \mathcal{D}_W(1; \theta, 1 - \delta), \quad C_W - 1 \leq \Lambda_W(1; \theta, 3/5) \quad (43)$$

the *common-threshold property*. This name only abbreviates the three displayed assertions.

Theorem 4.2 (Common-threshold bridge). *Let W have nonempty finite carriers, let $C_W \geq 3$ and $b \geq 1$, and let $0 < \theta$ and $0 < \delta \leq 1/64$. Assume (43). If $Q = \text{Qwidth}(b, \delta)$ and $Q\theta < 1/20$, then*

$$R^+ = \text{Int}_Q(W)^\top \quad (44)$$

satisfies

$$D(R^+) = C_W + \log_2 Q, \quad (45)$$

$$C_W + \log_2 Q \leq \text{Rob}(R^+; b, 3/5). \quad (46)$$

Moreover, $D(W) \leq Y_W$ implies

$$D(R^+) \leq Y_W + \log_2 Q. \quad (47)$$

The bridge is independent of the source graph and of the fourth-stage carrier definitions. Its proof, in Appendix B, is a defect induction on actual projections of retained column sets. The one-copy near-full estimate handles its base; inherited two- and three-copy consequences of the low profile handle the first increments; and the protocol's first real query drives the induction. A final

ceiling-safe count enters at the requested depth b . The selector protocol supplies the matching upper bound, so the conclusion is an equality.

For the canonical schedule, $Q = Q_{\text{br}}$ and $Q_{\text{br}}\theta = 1/160 < 1/20$. Thus the bridge output is definitionally the R^+ in (30), not a similarly sized replacement.

4.2 The complete one-level statement

Define the level overhead

$$H = a + \log_2 q_2 + \log_2 s + \log_2 q_{\text{al}} + \log_2 Q_{\text{br}}. \quad (48)$$

All three widths are positive powers of two in the canonical schedule.

Corollary 4.3 (One-level amplifier). *The successor R^+ is fixed from the public seed, source graph, parameters, and balanced-family entries before the source promise is used.*

(i) *If the graph is properly three-colourable and $D(R) \leq Y$, then*

$$D(R^+) \leq Y + H. \quad (49)$$

(ii) *If the graph is in the negative source case and the seed and scalar conditions hold, then*

$$D(R^+) = D_0 + H + 1, \quad (50)$$

$$D_0 + H + 1 \leq \text{Rob}(R^+; b_+, 3/5). \quad (51)$$

(iii) *If $m_N = |\text{Rows}(N)|$ and $n_N = |\text{Cols}(N)|$, the actual successor has*

$$|\text{Rows}(R^+)| = (q_{\text{al}} m_N)^{Q_{\text{br}}}, \quad (52)$$

$$|\text{Cols}(R^+)| = Q_{\text{br}} n_N^{q_{\text{al}}}. \quad (53)$$

On the positive source branch, the colouring removes the final template/source distinguishing bit from the raw upper protocol. The two interlace selector protocols then give (49). On the negative branch, the protected and scale-aligned certificates followed by Theorem 4.2 give exactly one additional bit. The carrier equations follow directly from (9) and canonical relabelling.

5 Iteration and the same-matrix size bound

Fix one formula Φ , one source branch S , and

$$b = \lceil \log_2(n(\Phi) + 2) \rceil. \quad (54)$$

The source branch remains fixed throughout the recursion.

5.1 A robust full-cube seed

Let $a_{\text{seed}}(b)$ be the least $a \geq 3$ such that

$$64(b + 1) \leq 2^a - 2, \quad (55)$$

and put $q_{\text{seed}} = 2^{a_{\text{seed}}(b)} - 2$. Define

$$F_b(x, i) = x(i), \quad x : [q_{\text{seed}}] \rightarrow \{0, 1\}, \quad i \in [q_{\text{seed}}]. \quad (56)$$

Its rows are all Boolean vectors and its columns are coordinates. If $y_0 = a_{\text{seed}}(b) + 1$, then

$$D(F_b) = y_0, \quad y_0 \leq \text{Rob}(F_b; b, 3/5), \quad w(F_b) = q_{\text{seed}}. \quad (57)$$

For the upper bound, Bob identifies the coordinate and Alice sends its bit. For the lower bounds, the unit-vector rows give an identity submatrix in the one-output indicator. In a row set of density 2^{-b} , the Boolean vectors span a real space of dimension at least $q_{\text{seed}} - b$. After retaining a $3/(5 \cdot 2^r)$ fraction of columns, the resulting indicator rank exceeds $2^{a_{\text{seed}} - r - 1}$ for $r = 0, 1, 2$, by (55). The fixed-output rank bound proves the three rungs in (57).

5.2 One matrix-and-budget recursion

At level i , the state consists of the actual current matrix $R_i : [m_i] \times [n_i] \rightarrow \{0, 1\}$ and a positive-branch budget y_i . Initialize R_0 with the canonical zero-based presentation of F_b and the budget above. Feed the level schedule the exact values

$$D_i = y_i + i, \quad Y_i = y_i, \quad b_0 = b_+ = b. \quad (58)$$

The schedule also uses the promise-independent moderate exponent

$$h(n) = \max \left\{ 5, (\log_2^\downarrow (\log_2^\downarrow n))^4 \right\}, \quad (59)$$

where the integer floor logarithm is total and is zero at zero. Let H_i be the level overhead from (48). Define

$$R_{i+1} = \text{Succ}(S, R_i; \pi_i), \quad y_{i+1} = y_i + H_i. \quad (60)$$

This is one dependent recursion. There is no separate positive matrix sequence, negative matrix sequence, freely chosen width list, or supplied success certificate.

Theorem 5.1 (Gap induction). *For every $0 \leq i \leq k$, once the uniform cutoff conditions hold,*

$$S \text{ is properly three-colourable} \implies D(R_i) \leq y_i, \quad (61)$$

$$S \text{ is in the negative source case} \implies \begin{cases} D(R_i) = y_i + i, \\ y_i + i \leq \text{Rob}(R_i; b, 3/5). \end{cases} \quad (62)$$

Proof. At level zero, both implications follow from (57). For the successor, the positive induction hypothesis and (49) give $D(R_{i+1}) \leq y_i + H_i = y_{i+1}$. On the negative branch, (58) turns the induction hypothesis into the exact seed assumptions needed at level i . Equations (50)–(51) give

$$D(R_{i+1}) = (y_i + i) + H_i + 1 = y_{i+1} + i + 1$$

and regenerate the robust property at depth b . The same matrices and budgets occur in both arguments. $\square$

5.3 Exact carriers of the actual recursion

At one level, let

$$\begin{aligned} x_1 &= q_1 m_i, & y_1 &= n_i^{q_1 + s + 1}, \\ x_2 &= q_2 y_1, & y_2 &= N_2, \\ x_3 &= s y_2, & y_3 &= x_2^s, \end{aligned}$$

and abbreviate

$$\begin{aligned} T &= x_3, & V &= 4s v(G)m_i, & B_0 &= 4s(s-3)m_i, \\ X &= T + V + B_0, & S_0 &= y_3, & D' &= x_2, & M_0 &= n_i^{s+1}S_0. \end{aligned}$$

Collecting the labelled multiplicities from (26)–(27) gives the exact identities

$$|\text{Rows}(A_i)| = 2^{22}XTVB_0, \quad |\text{Cols}(A_i)| = 184M_0S_0D'. \quad (63)$$

Because $N_i = A_i^\top$ and because the two later wrappers are literal interlaces, the actual successor satisfies

$$m_{i+1} = (q_{\text{al},i}|\text{Cols}(A_i)|)^{Q_{\text{br},i}}, \quad (64)$$

$$n_{i+1} = Q_{\text{br},i}|\text{Rows}(A_i)|^{q_{\text{al},i}}. \quad (65)$$

These are identities for R_{i+1} itself.

Suppose a number $z \geq 1$ bounds the ceiling logarithm of each of $X, T, V, B_0, M_0, S_0, D'$ and also satisfies $q_{\text{al}}, Q_{\text{br}} \leq z$. Then

$$|\text{Rows}(A)| \leq 2^{22+4z}, \quad |\text{Cols}(A)| \leq 2^{8+3z}, \quad (66)$$

and (64)–(65) give

$$w(R_{i+1}) \leq A(z) := \max\{1, (4z+8)z, z + (4z+22)z\} = 4z^2 + 23z \leq 27z^2. \quad (67)$$

Choose one branch-uniform initial bound z_0 for the seed, every ingredient, and every wrapper width through the fixed k levels, and define $z_{i+1} = A(z_i)$. Induction using the exact carrier equations proves

$$w(R_{i,j}) \leq z_i \quad \text{for every source-list index } j \text{ and every } i \leq k. \quad (68)$$

5.4 One cutoff before the formula

Appendix C gives the canonical schedule and the finite inequalities it must satisfy. For fixed k , all schedule widths, ingredient logarithms, and the scalar sequence in (67) are bounded uniformly over the source list by

$$Z_k(n) = 2^{A_k(1+\ell(n))^{d_k}}, \quad \ell(n) = \log_2^\downarrow(\log_2^\downarrow n), \quad (69)$$

for constants A_k, d_k . A polynomial in $\log \log n$ is eventually smaller than $\log n$, so for every fixed positive A, d, C there is a cutoff beyond which

$$C 2^{A(1+\ell(n))^d} \leq n. \quad (70)$$

Applying this with $C = 2k$ and enlarging the cutoff to cover the source and all structural and scalar gates yields one $U_k \geq \max\{k, n_{\text{src}}\}$, independent of the formula and branch, for which

$$z_k \leq \left\lfloor \frac{n}{2k} \right\rfloor = e(n, k). \quad (71)$$

Finally, if $w(M) \leq e(n, k)$, then both sides of M have size at most $2^{e(n,k)}$. Their product is at most $2^{2e(n,k)}$, and

$$2 \left\lfloor \frac{n}{2k} \right\rfloor \leq \left\lfloor \frac{n}{k} \right\rfloor \quad (72)$$

because $k > 0$. This proves the side and entry bounds for the same matrix $R_{k,j}$ controlled by the gap induction.

6 Assembly and verification boundary

We now prove [Theorem 1.1](#). Fix the two external contracts and $k \geq 1$. Choose the common cutoff U_k from [Section 5](#), enlarged so that $e(n, k) \geq 1$ beyond it. This choice depends on k and the two fixed contracts, but not on a formula or a source-list index.

Let Φ be a valid formula with $n(\Phi) \geq U_k$. The source theorem gives one nonempty ordered list $\mathcal{S}(\Phi)$. For every $j \in [t(\Phi)]$, run the single recursion [\(60\)](#) on the corresponding source branch and define

$$M_{\Phi,k,j} = R_{k,j}, \quad Y_{\Phi,k,j} = y_{k,j}. \quad (73)$$

The cutoff and exact carrier equations prove that every carrier is positive, so the structural fallback is never selected. This entire family is now fixed.

If Φ is satisfiable, source clause (S4) supplies an index j whose graph is properly three-colourable. The positive half of [Theorem 5.1](#) gives

$$D(M_{\Phi,k,j}) = D(R_{k,j}) \leq y_{k,j} = Y_{\Phi,k,j}.$$

If Φ is unsatisfiable, source clause (S5) places every branch in the negative source case. The negative half of the same theorem gives, for every j ,

$$D(M_{\Phi,k,j}) = D(R_{k,j}) = y_{k,j} + k = Y_{\Phi,k,j} + k.$$

This is an equality: the explicit selector protocols supply the upper bound at every level, while the protected, aligned, and bridged certificates supply the matching lower bound.

For every j , independently of the formula case, [\(68\)](#) and [\(71\)](#) give

$$w(M_{\Phi,k,j}) = w(R_{k,j}) \leq e(n(\Phi), k).$$

The conversion following [\(72\)](#) gives the two side bounds and the entry bound for this very $R_{k,j}$. The source list is nonempty, so none of the quantifiers is vacuous. This completes the proof.

6.1 A concise adversarial audit

The proof has been organized to make the following possible mistakes visible.

- (A1) The two external contracts precede k ; the cutoff precedes the formula; and the output family precedes both source implications.
- (A2) No colouring, negative source certificate, protocol lower bound, or final size assertion is an argument of a constructor.
- (A3) Every transpose swaps the two retention densities. The three profile rungs use column densities $(3/5)/2^r$ and debts $r = 0, 1, 2$.
- (A4) N_{ref} is the literal base-row restriction of N and retains all columns. The proof does not replace it by an abstract matrix with the same cardinalities.
- (A5) The negative result is exact because each lower bound is paired with an explicit selector upper protocol.
- (A6) The matrix chain is the recursion [\(60\)](#), and the width induction uses its exact carrier identities [\(64\)](#)–[\(65\)](#).

(A7) The root has exactly two outside mathematical inputs. The expander used upstream in one proof of the source theorem is not a third input to this construction.

The mathematical argument above is complete relative to those two stated inputs. A proof-assistant completion is a separate claim: it requires the finite root to compile, every dependency to be free of placeholders, and an axiom report containing only ordinary logical principles and the approved external interfaces. A successful document build by itself does not make that claim.

A The protected and scale-alignment frontiers

This appendix expands the only construction-specific lower-bound arguments. All row and column sets below are literal subsets of the matrices defined in [Section 3](#).

The proof invokes the inherited extension theorem in only four ways. Its parameters record the number v of coupled streams, a family width K , a retained mass h , and the number of communication bits already charged. The four substitutions are fixed before any protocol is inspected:

Call	Purpose	v	(K, h)	charged bits
(i)	a protected stop after Alice speaks	1	(κ, x_{Dom})	1
(ii)	a terminal pair with one ordinary and one foreign stream	2	$(\kappa, 49/64)$	1
(iii)	protected donor mass that stops when Bob first speaks	2^{e-1}	$(s\kappa q_2, \rho_1 2^{-(2^e)-1})$	0
(iv)	the exceptional terminal half-block	$2^{e_{\text{exc}}-1}$	$(2sq_2, \rho_0 2^{-(2^{e_{\text{exc}}})-1})$	0

Here κ is the actual Stage-2 certificate width and x_{Dom} is the displayed dominant protected-row fraction. In (iii), the selected dyadic donor class has width 2^e with $e \geq 1$; in (iv), the canonical exceptional class has width $2^{e_{\text{exc}}}$ with $e_{\text{exc}} \geq 2$. Thus the table uses literal half-blocks of the displayed cardinalities. The public scalars ρ_1 and ρ_0 are the retained fractions after, respectively, the early and terminal loss charges. For every row, the carrier maps are checked to land in the current matrix, (B1), (B2), or (B3) supplies the balanced-family count, the scalar checklist supplies the retained mass, and the incoming seed rung supplies the charged bits.

A.1 Protected Stage 4

Fix a retained restriction of N keeping a $(1 - \eta_R)$ fraction of rows and a $(1 - \eta_C)$ fraction of columns, and suppose that a protocol for it has cost below C . Because the rows and columns of $A = N^T$ were duplicated into labelled reservoirs, deletion damage can be charged by reservoir. The protected-loss inequalities imply:

- (P1) all but a $1/1000$ fraction of template rows keep enough labels;
- (P2) all but a $\delta_V/16$ fraction of vertex-source rows keep enough labels;
- (P3) all but a $1/(16(s-3))$ fraction of blocker-source rows keep enough labels;
- (P4) every selector and diagonal column class required by the proof has a surviving representative.

These are integer counts obtained by clearing positive denominators.

The ordinary balanced-family count then leaves a set of good Stage-2 coordinates on which the singleton, pair, and reference patterns required by the proof all occur. If too many coordinates were

bad, the missing family members would exceed the total deletion allowance. Projection covers make the exceptional set finite and give the bound $e_{\max}$ used by the schedule.

Follow the alleged protocol while keeping actual surviving template rows. At a reached node, remember the compatible row and column sets, represented good coordinates, their selector and diagonal representatives, the partial source colouring fixed so far, and a debt in $\{0, 1, 2\}$ recording which profile rung is being used. At an Alice node, her actual query partitions all row reservoirs; conservation of labels selects a nonempty child with the required mass. At a Bob node, his actual query partitions columns; a selector or diagonal representative either survives in a child or spends one unit of profile debt.

If this routing could not synchronize a represented coordinate, its answers would define a three-colouring of G with fewer than $\gamma e(G)$ monochromatic edge occurrences. Vertex rows enforce consistency at the two incidence positions, blocker rows keep the unused colours distinct, and

$$\frac{64s(2^a - 2)}{\gamma/24} \leq q_2 \quad (74)$$

ensures that the discarded coordinates account for less than a γ fraction of edge occurrences. This contradicts the negative source condition.

Stop at the first Bob node or when all selector bits of a represented coordinate have been exposed. The stopping routes are prefix-free, their row sets are disjoint, and Kraft's inequality bounds their total binary capacity by one. The weighted frontier count therefore produces a stop with enough template and source-owned mass. If all selector bits have been exposed, the surviving selector and diagonal columns embed the appropriate incoming seed rung. If Bob speaks first, the protected one-stream extension lemma embeds a hard restriction in one of his real children. Either suffix requires at least $C - d$ further bits after a route of length d , contradicting the alleged protocol. This proves (34).

For the reference rungs, retain only the base-labelled raw columns that form the rows of N_{ref} . At rung $r \in \{0, 1, 2\}$, the exact template density and the ceiling gate in (113) leave at least $9s/(16 \cdot 2^r)$ actual Stage-3 blocks. The physical reference map sends these rows and all retained columns into the literal restriction of N_{ref} . The fixed reference extension theorem embeds the appropriate seed rung and gives cost at least $C - 1 - r$. This proves (36).

A.2 The alignment frontier

Consider the literal retained game on $U = \text{Int}_{q_{\text{al}}}(N)$ with row density $1 - \delta_U$ and column density θ . After removing coordinates with excessive physical or reference-row loss and those excluded by the three projection covers, obtain an exceptional set E_0 of size at most $e_{\max}$. Each coordinate outside E_0 has the initial reference floor and every column projection needed below.

Assume a protocol Q of cost at most $C_W - 1$. Follow only actual Alice questions until Bob first speaks or until $r_{\text{al}} = \log_2 q_{\text{al}}$ Alice answers have been seen. The first-Bob routes and final length- r_{al} route tokens form a prefix-free set $\mathcal{T}$. If d_v is the route length and $u_v = 2^{r_{\text{al}} - d_v}$ its residual dyadic capacity, then

$$R_U = \bigsqcup_{v \in \mathcal{T}} \mathcal{R}_v, \quad \sum_{v \in \mathcal{T}} 2^{-d_v} \leq 1, \quad \text{cost}(Q_v) \leq C_W - 1 - d_v. \quad (75)$$

For each clean starting coordinate, route a reference certificate through the Alice queries. If its threshold at remaining depth k is $A_k = 2^k A_0$, at least one child contains A_{k-1} reference rows; keep both children when both qualify and otherwise keep the unique qualifying child. Stop these descendants at the frontier. A certificate remembers its physical coordinate, stopping token, actual

reference-row subset, and whether it split. Protected rows from the same physical coordinate are counted independently: they may reach several stopping tokens.

At a first-Bob node, small residual capacity is contradicted by one of the three reference rungs. Large residual capacity selects a dyadic class of donors. Product clause (B1), the actual carrier maps, and the fixed multi-stream extension lemma construct a hard restriction in a real child. Thus only a bounded amount of protected mass associated with clean certificates can stop early.

A.3 Owners, donors, and terminal mass

We spell out the terminal count because confusing a certificate route with physical mass would invalidate the proof. Put $\vartheta = 9/16$. Let J_{clean} be the clean certificates ending at final tokens, let $c = |J_{\text{clean}}|$, let n_T be the number of final tokens, and put $K_B = \sum_{v \text{ first Bob}} u_v$. Certificate conservation and Kraft's inequality give

$$c \geq q_{\text{al}} - \frac{e_{\max}}{1 - \vartheta}, \quad K_B + n_T \leq q_{\text{al}}. \quad (76)$$

Normalize protected row mass by $m = |\text{Rows}(N)|$. If $P_v(J)$ is the mass from the coordinates of certificates in J reaching a first-Bob token v , the contrapositive of the heavy-node extension gives

$$P_B := \sum_{v \text{ first Bob}} P_v(J_{\text{clean}}) < 128\mu B_{\text{pair}} L_q s K_B \leq \frac{128\mu B_{\text{pair}} L_q s e_{\max}}{1 - \vartheta}. \quad (77)$$

Call a certificate Bob-heavy if its early mass exceeds $\eta_R/4$. Markov counting and (77) give

$$e_B \leq \frac{512\mu B_{\text{pair}} L_q s e_{\max}}{(1 - \vartheta)\eta_R} \leq E_B. \quad (78)$$

The remaining certificates are the good donors J_{good} .

The owner $o(j)$ of a good donor is the final token at which its clean certificate stops. Final-reference uniqueness makes o injective. The exceptional final tokens are those outside $o(J_{\text{good}})$; if their count is e_T , then

$$e_T \leq \left\lceil \frac{e_{\max}}{1 - \vartheta} \right\rceil + E_B = E_T. \quad (79)$$

The schedule ensures $E_T \leq q_{\text{al}}/8$, and therefore $|J_{\text{good}}| \geq 7q_{\text{al}}/8 > q_{\text{al}}/2$.

For a donor j and final token T , let $m_{j,T}$ be the normalized physical protected mass from j that reaches T . Mass missing from the owner is partitioned into initially deleted rows, rows lost from the unique clean certificate route, rows stopped at first-Bob tokens, and rows reaching other final tokens. The first three charges are at most $\eta_R/16$, strictly below $\eta_R/16$, and at most $\eta_R/4$, respectively. If the last charge were at most $\eta_R/2$, the owner would retain more than $1 - \eta_R$ of its physical copy of N and a $1 - \eta_C$ column projection. The protected theorem would give residual cost at least C , while a final token has at most $C - 1$. Hence

$$\sum_{T \neq o(j)} m_{j,T} > \frac{\eta_R}{2} \quad (j \in J_{\text{good}}). \quad (80)$$

It remains to control transfers to other good owners. The ordinary baseline occupies the fraction $g = 91/92$ of one physical copy. Let ε_i be its deficit at $o(i)$ and define

$$\lambda_0 = q_2^{1-b_1}, \quad \delta_i = \lambda_0 + 5\varepsilon_i + \frac{2}{q_2}. \quad (81)$$

Disjoint owner baselines give

$$\sum_{i \in J_{\text{good}}} \varepsilon_i \leq \frac{\eta_R}{8g} < \frac{1}{5}. \quad (82)$$

For distinct good donors j, i , if $m_{j,o(i)} \geq 2\mu\delta_i$, product clause (B2) and the two-stream extension construct a literal restriction at $o(i)$ of cost at least C . This contradicts its $C - 1$ suffix budget, so

$$m_{j,o(i)} < 2\mu\delta_i. \quad (83)$$

The three owner-transfer gates in (115) and (82) now yield

$$\sum_{\substack{i \in J_{\text{good}} \\ i \neq j}} m_{j,o(i)} < 2\mu \sum_i \delta_i \leq \frac{3\eta_R}{16} < \frac{\eta_R}{4}. \quad (84)$$

Combining this with (80), every good donor sends more than $\eta_R/4$ of its mass to exceptional final tokens.

If there are no exceptional tokens, this is impossible. Otherwise, a double pigeonhole chooses one exceptional token T_* , one of the two protected reservoir types, and a donor set J_* satisfying

$$|J_{\text{good}}| \leq 2e_T |J_*|, \quad \delta_T \leq \frac{\eta_R}{8\mu e_T} \leq \text{density}_{T_*}(j) \quad (j \in J_*). \quad (85)$$

The terminal gates

$$4E_T u_T B_{\text{pair}} < q_{\text{al}}, \quad u_T \delta_T \geq 8s \quad (86)$$

leave u_T rich donors after projection-cover loss. Choose a literal half-block of size $u_T/2$. Product clause (B3) and the exceptional-terminal extension construct a restriction of the actual suffix at T_* with cost at least C , contradicting its cost at most $C - 1$. This completes the proof of the scale-aligned certificate.

B Defect induction for the generic bridge

We prove [Theorem 4.2](#). Expanding the low-profile part of (43) and applying the inherited finite two-copy ladder gives

$$C_W \leq \Lambda_W(2; 2\theta, (3/5)^2). \quad (87)$$

Because $4\theta < 1$, the inherited three-copy row step gives

$$C_W + 1 \leq \mathcal{D}_W(3; 4\theta, (3/5)^2). \quad (88)$$

These two estimates and the near-full estimate are the bridge's only communication lower-bound inputs.

Write $m = |\text{Rows}(W)|$, $n = |\text{Cols}(W)|$, and

$$\lambda = -\ln(1 - \delta), \quad A_a = \lceil m2^a \theta \rceil. \quad (89)$$

For a nonempty $K \subseteq \text{Cols}(W)^p$, define its logarithmic defect

$$\text{Def}_p(K) = \ln \frac{n^p}{|K|}. \quad (90)$$

Projecting K to some p' of its p coordinates can be done with defect at most $(p'/p) \text{Def}_p(K)$. If the coordinates are partitioned into two sets, the sum of the two projection defects is at most $\text{Def}_p(K)$. Both facts follow from finite product counting and use actual projections of K .

For $p \geq 2$ let

$$B(p) = \left(1 + \frac{6}{5p}\right) \ln 2. \quad (91)$$

The elementary inequalities needed in the induction are

$$\begin{aligned} B(p) &\geq \ln 2, & B(p') &\geq B(p) \quad (2 \leq p' \leq p), \\ B(\lceil p/2 \rceil) &\geq \frac{\lceil p/2 \rceil}{p} (B(p) + \ln 2), & \frac{3}{p} B(p) &\leq \frac{7}{5} \ln 2. \end{aligned} \quad (92)$$

For $a \geq 0$ and $p > 2^{a-1}$, with $2^{-1} = 1/2$, put

$$c_a(p) = p - 2^{a-1}. \quad (93)$$

Lemma B.1 (Defect induction). *Assume*

$$3\lambda + \frac{7}{5} \ln 2 \leq -2 \ln(3/5). \quad (94)$$

Let $a \geq 0$, $p > 2^{a-1}$, and $2^a \theta \leq 1$. Every restriction of $\text{Int}_p(W)$ that keeps at least A_a rows in each active copy and has nonempty column set K satisfying

$$\text{Def}_p(K) \leq c_a(p) \lambda \quad (95)$$

has communication complexity at least $C_W + a$.

The proof uses the following one-bit-down auxiliary statement: for $a \geq 2$, the relaxed condition

$$\text{Def}_p(K) \leq c_a(p) \lambda + B(p) \quad (96)$$

implies cost at least $C_W + a - 1$.

At $a = 2$, project to three copies. Its defect is at most $3\lambda + (3/p)B(p)$, hence at most $-2 \ln(3/5)$ by (92) and (94). The column density is therefore at least $(3/5)^2$, and (88) gives cost at least $C_W + 1$.

For the auxiliary induction step, consider the first real protocol bit. If Alice speaks, assign each active copy to a child retaining at least half its rows; since $A_{a-1} \leq \lceil A_a/2 \rceil$, that child has the next row floor. If p_i and L_i are the number of copies and projection defect assigned to child i , then $p_0 + p_1 = p$ and $L_0 + L_1 \leq \text{Def}_p(K)$. The inequality

$$(p_0 - 2^{a-2})_+ + (p_1 - 2^{a-2})_+ \geq p - 2^{a-1}$$

forces a real child satisfying the auxiliary hypothesis at level $a - 1$. If Bob speaks, keep a child containing at least half of K , increasing defect by at most $\ln 2$, and project to $p' = \lceil p/2 \rceil$. Direct substitution and (92) give

$$\frac{p'}{p} c_a(p) \leq c_{a-1}(p'), \quad \frac{p'}{p} (B(p) + \ln 2) \leq B(p').$$

The auxiliary induction hypothesis again applies to a real child. Adding the first bit proves the auxiliary statement.

For the main lemma, the case $a = 0$ projects to one copy with defect strictly below λ , so the column density exceeds $1 - \delta$ and the near-full common-threshold estimate gives C_W . At $a = 1$, an Alice first bit reduces to $a = 0$ by the same split. A Bob first bit keeps half the columns and projects to two copies; its defect is at most $2\lambda + \ln 2 \leq -2 \ln(3/5)$, so (87) applies. For $a \geq 2$, an Alice first bit yields the main induction hypothesis in one child; a Bob first bit yields the auxiliary statement at the same level. In both cases the parent costs at least $C_W + a$. This proves [Theorem B.1](#).

B.1 Entering the induction at the requested depth

Put $Q = Q_{\text{width}}(b, \delta)$, $r_Q = \log_2 Q$, and $L = b \ln 2$. Since $\lambda \geq \delta$ and $80b \leq Q\delta$,

$$Q \geq \frac{80L}{\lambda}, \quad Q \geq 80. \quad (97)$$

The bound $\delta \leq 1/64$ implies the fixed margin (94).

Let d be the least positive integer for which $(1 - \delta)^d \leq 2^{-b}$. Its minimality says

$$(d - 1)\lambda < L \leq d\lambda. \quad (98)$$

For $t = 0, 1, 2$, set

$$a_t = r_Q - t, \quad p_t = 2^{a_t-1} + d. \quad (99)$$

Then $p_t > 2^{a_t-1}$ and $p_t \leq Q$. The ceiling-safe good-copy count is

$$(p_t - 1)m + (Q - p_t + 1)(A_{a_t} - 1) < Qm \frac{3/5}{2^t}. \quad (100)$$

After division by Qm , the left side is strictly below

$$2^{-t-1} + \frac{L}{Q\lambda} + \frac{Q\theta}{2^t} < \frac{3/5}{2^t},$$

using $L/(Q\lambda) \leq 1/80$ and $Q\theta < 1/20$.

Take any retained member in the t -th rung for $R^+ = \text{Int}_Q(W)^\top$ and transpose it. Its row count and (100) leave at least p_t outer copies with A_{a_t} rows. Project its retained columns to those copies. The column density is still at least 2^{-b} , so its defect is at most L . Moreover,

$$2^{a_t}\theta = Q\theta/2^t < 1, \quad L \leq d\lambda = c_{a_t}(p_t)\lambda.$$

The defect induction gives cost at least $C_W + r_Q - t$. These three values are exactly the rungs in (46). The full-matrix $t = 0$ case gives the lower bound on $D(R^+)$, and the selector protocol gives the matching upper bound as well as (47).

C The finite schedule and its arithmetic gates

This appendix records one deliberately generous parameter schedule. Its purpose is to make the construction total and every comparison finite. No number is selected from a colouring or a negative source certificate.

C.1 The schedule

Besides the current public matrix state, let the schedule read $I = (D, Y, b_0, b_+)$ and a natural exponent h . Define

$$\begin{aligned} \sigma &= \max\{6, h + 2\}, & s &= 2^\sigma, \\ r_2 &= \lceil \log_2(12se(G)) \rceil, & q_2 &= 2^{r_2}, \\ \ell_1 &= \lceil \log_2(r_2 + 18) \rceil, & u_1 &= \ell_1 + 2, \\ a &= h + 2 + 16u_1^2, & q_1 &= 2^a - 2, \\ b_1 &= 2^{u_1}, & b_2 &= \lceil \log_2(32sq_1m_R + 2) \rceil + 6. \end{aligned} \quad (101)$$

Put

$$\mu = \frac{1}{184}, \quad \zeta = \frac{\gamma}{24}, \quad \delta_V = \frac{\gamma e(G)}{2\Delta v(G)}, \quad (102)$$

and then

$$\eta_C = \frac{1}{4} \min \left\{ \frac{2^{-21}}{1000s}, \frac{2^{-22}\delta_V}{16}, \frac{2^{-22}}{16(s-3)} \right\}, \quad (103)$$

$$\eta_R = \frac{1}{4} \min \left\{ \frac{\mu}{100}, \frac{\mu\zeta}{32s}, \eta_C \right\}, \quad n_\eta = \max \{2, \lceil 1/\eta_R \rceil\}. \quad (104)$$

For the reference and alignment widths, set

$$\begin{aligned} U_{\text{ref}} &= \max \{16, D + Y + b_0 + b_+ + r_2 + \sigma + a + n_\eta + 64\}, \\ L_{\text{ref}} &= \lceil \log_2 U_{\text{ref}} \rceil, \\ k_{\text{ref}} &= 2^{20} + 4096(L_{\text{ref}} + 1)^2, \\ r_{\text{al}} &= 64(k_{\text{ref}} + L_{\text{ref}} + \sigma + 1), \\ q_{\text{al}} &= 2^{r_{\text{al}}}, \\ \delta_U &= \frac{\eta_R}{16q_{\text{al}}}, \\ \theta &= \frac{1}{160Q_{\text{br}}}. \end{aligned} \quad \begin{aligned} b_T &= 2r_{\text{al}} + L_{\text{ref}} + 32, \\ Q_{\text{br}} &= \text{Qwidth}(b_+, \delta_U), \end{aligned} \quad (105)$$

Let $B_{\text{ref}}, B_{\text{sing}}, B_{\text{pair}}$ be the least finite projection-cover exponents needed, respectively, for the small-reference, single-coordinate, and paired-coordinate calls at column density θ . Define

$$\begin{aligned} t_{\text{ref}} &= 2^{k_{\text{ref}}}, \\ e_{\text{max}} &= (B_{\text{ref}} - 1)(t_{\text{ref}}/2 + \log_2 t_{\text{ref}} - 2) + (B_{\text{sing}} - 1) + 2(B_{\text{pair}} - 1), \\ L_q &= \lceil \log_2(2q_{\text{al}}) \rceil + 1, \\ E_B &= \left\lceil \frac{512\mu B_{\text{pair}} L_q s e_{\text{max}}}{(1 - 9/16)\eta_R} \right\rceil, \\ E_T &= \left\lceil \frac{e_{\text{max}}}{1 - 9/16} \right\rceil + E_B, \\ \delta_T &= \frac{\eta_R}{8\mu \max\{1, E_T\}}, \\ u_T^* &= \max \{1, \lceil 8s/\delta_T \rceil\}, \end{aligned} \quad u_T = 2^{\lceil \log_2 u_T^* \rceil}. \quad (106)$$

Finally,

$$\begin{aligned} L_h &= \max\{b_2, b_T\} + \max\{q_{\text{al}}, u_T\} + 4, \\ H_{\text{max}} &= L_h + 8q_1 + r_{\text{al}} + r_2 + (D + a) + 10, \\ L_2 &= \max \{2, \lceil \log_2(H_{\text{max}} + 2) \rceil\}, \\ k_2 &= a + h + 4096L_2^2 + 8, \end{aligned} \quad t_2 = 2^{k_2}. \quad (107)$$

This supplies exactly the six public parameters in (18). If the source and seed carriers are positive, $D \geq 3$, $b_+ \geq 1$, $h \geq 5$, and

$$k_2 + 2 \leq r_2, \quad (108)$$

then the structural conditions in (19) hold. In particular, $4t_2 = 2^{k_2+2} \leq 2^{r_2} = q_2$, the guarded ranks are below their displayed width, and all three wrapper widths are positive powers of two.

C.2 Five formula-uniform comparisons

Structural readiness alone does not imply the five quantitative comparisons used by the negative proof. The uniform formula cutoff proves them separately:

$$m_{\text{HS}} \leq h, \quad \frac{64sq_1}{\zeta} \leq q_2, \quad u_T \leq q_{\text{al}}, \quad a + 2 + \sigma \leq b_0, \quad 100(V_0 + B_0) \leq T_0. \quad (109)$$

Here m_{HS} is the fixed inherited hard-seed threshold and T_0, V_0, B_0 are the actual template, vertex, and blocker reservoir counts. The balanced count on t_2 coordinates makes every $(q_1 m_R)^{t_2}$ pattern occur, so $N_2 \geq (q_1 m_R)^{t_2}$; this is the lower bound used for the final reservoir-dominance comparison.

Together with the schedule definitions, the five rows imply the following groups of gates.

Raw upper protocol. The schedule makes s a positive power of two and proves

$$s = 2^{\log_2 s}, \quad s \leq 2^a. \quad (110)$$

Protected losses.

$$\begin{aligned} 8q_1 &\leq t_2, & \frac{64sq_1}{\zeta} &\leq q_2, \\ \eta_R &< \frac{(1/4)\mu\zeta}{8s}, & \eta_C &\leq \frac{1}{2} \min \left\{ \frac{2^{-21}}{1000s}, \frac{2^{-22}\delta_V}{16}, \frac{2^{-22}}{16(s-3)} \right\}. \end{aligned} \quad (111)$$

Reference blocks. The schedule gives

$$1 \leq k_{\text{ref}}, \quad m_{\text{HS}} + 2 \leq \log_2 s < b_T, \quad k_{\text{ref}} + 2 \leq \log_2 q_{\text{al}}, \quad q_{\text{al}} 2^{-b_T} \leq \frac{4}{25025}, \quad a + 2 + \log_2 s \leq b_0. \quad (112)$$

Let $w_j = s/2^j$, $T_j = 9w_j/16$, $d_j = w_j 2^{-b_T}$, and

$$\beta_j = \frac{(3/5)/2^j + \tau_{\text{act}} - 1}{\tau_{\text{act}}}, \quad \tau_{\text{act}} = (1 - 2^{-20}) \frac{T}{T + V + B_0} + 2^{-22}.$$

For $j = 0, 1, 2$, the exact ceiling gate is

$$T_j \leq \left\lceil \frac{s(\beta_j - d_j)}{1 - d_j} \right\rceil. \quad (113)$$

The ceiling covers the whole quotient. This is the extremal-bin count used for the three actual reference rungs.

Bridge and owner transfer. The definitions give

$$\delta_U q_{\text{al}} = \eta_R/16, \quad 80b_+ \leq Q_{\text{br}} \delta_U, \quad Q_{\text{br}} \theta = 1/160 < 1/20, \quad \delta_U \leq 1/64. \quad (114)$$

With $g = 91/92$ and $\lambda_0 = q_2 2^{1-b_1}$, the three owner-transfer comparisons are

$$2\mu q_{\text{al}} \lambda_0 \leq \frac{\eta_R}{16}, \quad \mu \leq \frac{g}{20}, \quad \frac{64\mu q_{\text{al}}}{\eta_R} \leq q_2. \quad (115)$$

They follow from the cleared natural comparisons $\eta_R^{-1} \leq q_{\text{al}}$, $4q_{\text{al}}^2 \leq q_2$, and $q_2^2 \leq 2^{b_1}$.

Exceptional terminal. The terminal width estimates give

$$8 \max\{1, E_T\} \leq q_{\text{al}}, \quad 4E_T u_T B_{\text{pair}} < q_{\text{al}}, \quad 4t_{\text{ref}} \leq q_{\text{al}}, \quad u_T \delta_T \geq 8s. \quad (116)$$

For example, if $K = k_{\text{ref}} + \sigma$, the schedule bounds $E_T \leq 2^{6K+51}$, $u_T \leq 2^{8K+58}$, and $B_{\text{pair}} \leq 2^{K+15}$, while $15K + 126 < r_{\text{al}}$. This proves the strict middle inequality without decimal rounding. Also $e_{\text{exc}} = \log_2 u_T \geq 2$ is fixed before an alleged protocol is inspected, exactly as required by product clause (B3).

C.3 Why one uniform cutoff exists

For fixed k , induct over the actual k -level state. Finite sums, products, maxima, and fixed powers of quantities bounded by $2^{A(1+\ell)^d}$ remain within another bound of the same form. Taking a ceiling logarithm makes the bound polynomial in $1 + \ell$. Every exponentiation in the schedule is applied either to such a logarithm or to a fixed polynomial in such logarithms. The balanced-family size bound (13) has the same closure property, and the exact carrier recurrence is controlled by the quadratic arrow (67).

Taking a maximum over the finite source list before using either source promise produces the branch-uniform envelope (69). The hard-seed row follows because $\ell(n)^4$ tends to infinity. The Stage-2 fit, protected source width, and input-depth rows reduce to a fixed polynomial in $1 + \ell(n)$ being at most $\log_2^\downarrow n$. The terminal definitions first bound the projection exponents and e_{max} , after which the deliberately larger alignment exponent gives $u_T \leq q_{\text{al}}$. The balanced-family lower count gives reservoir dominance. These are precisely the five rows in (109); all other gates above are their direct consequences.

The elementary comparison (70) therefore provides one cutoff that simultaneously proves structural legality, every negative scalar gate, positive carriers, and the final naming-width bound. It uses no monotonicity assumption about how the source list changes with the formula.

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